

# SWATHI ANUMANCHINENI

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## PROFESSIONAL HIGHLIGHTS

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- Data Scientist / Data Engineer with experience in Python, SQL, Machine Learning, and Deep Learning. Skilled in data preprocessing, model development, and visualization using Scikit-learn, TensorFlow, and Power BI. Strong foundation in ETL pipelines, data modeling, APIs, and deployment with Flask.

## EDUCATION

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**Hofstra University - Hempstead, New York**

**Master of Science in Data Science**

(Fall 2023 – Spring 2025)

- **Capstone Title:** Conversational AI for Mental Health Support -Empathy Through Technology.  
CGPA: 3.37/4
- **Advisor:** Prof. Andrew Lane

**Kavitha Memorial Degree College, KU - Khammam, Telangana**

**Bachelor of Science in Mathematics, Chemistry and Computer Science** (Fall 2019 – Spring 2022)

- *Achieved Highest Honors in the B.Sc. program in Mathematics, Chemistry, and Computer Science.*  
CGPA: 89%/100%

## TECHNICAL SKILLS

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- **Programming:** Python, SQL, R, MATLAB
- **Libraries/Frameworks:** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV, NLTK
- **Tools:** Jupyter, Colab, Power BI, RapidMiner, MySQL, Git, Excel
- **ML/DL:** Regression, Random Forest, Gradient Boosting, CNN, NLP, Time Series, Clustering
- **Data:** EDA, Feature Engineering, Data Augmentation, Batch Normalization, Dropout, PCA

## ACADEMIC PROJECTS

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**Hofstra University – Hempstead, New York**

**House Rent Prediction | Python, SQL, Scikit-learn**

- Built and optimized ML models (KNN, Random Forest, Linear Regression, Gradient Boosting) to predict house rents.
- Achieved  $R^2$  of 0.87 using Linear Regression; identified key features influencing rents.
- Applied SQL & Pandas for EDA and visualized insights to support decision-making.

**Digit Segmentation Using Deep Learning | TensorFlow, CNN, U-Net**

- Designed CNN-based model for digit segmentation from noisy images.
- Enhanced accuracy using grayscale preprocessing, data augmentation, and optimization strategies.
- Achieved high segmentation performance with IoU, precision, and recall metrics.

**Adult Employee Salary Classification | RapidMiner, SQL**

- Developed Logistic Regression and Decision Tree models to predict salaries ( $\leq 50K$  or  $> 50K$ ).
- Decision Tree achieved 85.47% accuracy; Logistic Regression excelled in  $\leq 50K$  prediction.
- Conducted EDA to uncover salary drivers such as age, capital gain, and work hours.

## Empathetic Conversational AI Chatbot | Python, Transformers, Flask

- Developed chatbot using GPT-2 & DialoGPT fine-tuned on Empathetic Dialogues dataset (64k entries)
- Performed EDA on dialogue complexity and emotion distribution.
- Built deployment pipeline with Flask API for real-time interaction in mental health contexts.

## INTERNSHIP

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### CJT Inc. – Technical & Data Analysis Intern

*May 2024 – Till date*

- Redesigned demographic modules with PHP, Bootstrap, and Highcharts, reducing training time by 25%.
- Optimized MySQL queries and dynamic filters, cutting page load time by 40%.
- Managed Power BI dashboards, decreasing ad-hoc report requests by 30%.
- Cleaned and analyzed 1,500+ caregiver records in Python, identifying underserved ZIP codes.
- Collaborated with analysts using Slack to align filters with ACS/Census data.

**Tech Stack:** PHP, HTML/CSS, JavaScript, Highcharts, Python, Jupyter, MySQL, Power BI.

## HOBBIES/LEISURE ACTIVITIES

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- **Sports** – Basketball, Running.
- **Hobbies & Interests:** Chess, Rubik's Cube, Sudoku.
- **Food** – Culinary interest in Indian, Thai and Pakistan cuisines.
- **Other Interests:** Dance, Agriculture (Organic Farming)